Hanna **CHETOUANE**

Master's student in Applied Mathematics - Scientific Computing and Mathematics of Innovation



I am currently a Master's 2 student in Scientific Computing and Innovation Mathematics, and I am interested in the mathematical modeling of physical phenomena. My Master's 1 internship allowed me to discover this field in more detail and made me curious and motivated to explore its various opportunities. I am currently looking for a 6-month end-of-studies internship in this field.

About me

Nationality : French
Birth : 01/16/2003
Adress : 13 rue du Maennelstein
Strasbourg, France

Soft skills

- Good communication and teamwork skills
- Good adaptability and observation skills
- Perseverance and ability to surpass myself

Interests

- Mathematics and computer science
- Figure skating
- Nature and environment

Langages

French	C2	Mother tongue
English	B1/B2	● ● ● ○ ○

0768186222
hanna.chetouane@etu.unistra.fr
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Curriculum

2024–today — **CSMI Master (Scientific Computing and Mathematics of Innovation)**
University of Strasbourg
Dual competence in Mathematics and Computer Science
Mathématiques appliquées: Modeling, simulation, optimization, uncertainty management and model reduction, data analysis and processing (signals, sounds, images)
Computer science: Programming, software development, high-performance computing, machine learning, artificial intelligence, devops, and project management

2021–2024 — **Bachelor of Applied Mathematics**
University of Strasbourg
Fundamental bases in analysis (integration, function sequences and series), algebra, Fourier series, differential calculus, probability and statistics, as well as computer science.

Hard skills

Programming

Python ***~5 modules*** - Data analysis, uncertainty quantification simulation, machine learning, AI
C++ ***~4 modules + projects*** - PDE modelling, numerical simulation, parallel computing
Latex - Reports and projects presentations
C ***~1 module*** - Operating system
Rust ***~1 module*** - Scientific computing, equation solving
Sql ***~1 module*** - Database management
R ***~1 module*** - Mathematical statistics

Tools

- **GitHub:** CI workflows, branch management, pull requests, team collaboration
- **CMake:** Automated compilation and unit testing (CTest, Catch2)
- **Docker:** Environment management and image creation
- **Paraview:** Visualization of simulations in 2D and 4D
- **GMSH:** Mesh generation
- **Feel++:** Finite Element C++ library for solving PDEs
- **Eigen:** C++ linear algebra library
- **JAX, PyTorch:** Machine learning and AI frameworks
- **MPI, Kokkos, OpenMP:** Parallel computing for CPU/GPU

Experiences

June–Aug 2025 (2 months) — **M1 internship at Avnir energy**
Developed a simulation tool in C++ using Feel++ to model fluid-structure coupling, enabling the simulation of acoustic wave propagation in a nuclear power plant tank following an impact on the wall.

2022–today — **Mathematics Support**
From middle school to post-baccalaureate, I help students with their homework and general understanding of their mathematics lessons.

2018–today — **Figure Skating Instructor**
I coach at the CSGSA ("Club des sports de glace de Strasbourg Alsace") at all levels, including Competitors, Baby Club, and Adults.

Degrees

2024 **Bachelor of Applied Mathematics** in University of Strasbourg

2021 **Baccalaureate** at Sainte-Anne high school, Strasbourg

Certificates

2024 **BF1** (first ice sports coaching certificate)

2024 **PSC1** (Level 1 prevention and first aid)

2022 **B driving license**